TabuLLM

Synthetic Data Augmentation Benchmark Report

LLM-Driven Tabular Data Generation
Fidelity & Downstream ML Evaluation
across Six Medical Classification Datasets

Run ID	run_e	**Date**	2026-05-21
LLM Model	google/gemma-4-e2b	**Provider**	Local (LM Studio)
Folds	10	**Strategy**	Random
Datasets	6 medical	**ML Models**	10 classifiers
Ctx %	20%	**Gen %**	80%

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Executive Summary

This report presents the complete results of the TabuLLM benchmark pipeline, evaluating the effect of LLM-driven synthetic data augmentation on downstream machine learning classification performance across six publicly available medical datasets.

Experimental protocol. Each dataset undergoes 10-fold cross-validation. Per fold: (i) the dataset is split 80/20 into train and test; (ii) 20% of the training split is fed as real-data context to the LLM; (iii) the LLM generates a synthetic complement equivalent to 80% of the training split; (iv) context + synthetic rows form the *augmented* training set; (v) both a *baseline* model (real data only) and an *augmented* model are trained and evaluated on the same test split, enabling a paired comparison. Final metrics are the mean ± standard deviation across the 10 folds.

Key Findings

- **Synthetic fidelity (JSD)** averaged **87.6%** across all datasets — distributions are closely preserved.
- **All 10 ML models improved** on every dataset; accuracy deltas ranged from +0.043 to +0.067, F1 deltas from +0.057 to +0.070.
- **Privacy risk** remained below 5% on all datasets while **novelty scores** exceeded 1.27, confirming diverse, non-memorised generation.
- **Data density** was 100% across the board — the LLM generated fully populated rows with no missing values.
- The **SVM** and **KNN** models recorded the highest average F1 delta (+0.0647 and +0.0632 respectively) across datasets.

1. Cross-Dataset Overview

1.1 Fidelity Metrics — Averaged Across All Datasets

Table 1.1 reports mean fidelity values computed by averaging each metric across the six datasets (each already a 10-fold mean).

Table 1.1: Global average fidelity metrics across all 6 datasets.

Metric	Global Mean	Cross-Dataset Std	Notes
Synthetic Fidelity (JSD)	87.56%	0.52%	Numerical distribution similarity
Correlation Stability	83.83%	0.80%	Multivariate correlation preservation
Distribution Consistency	86.35%	0.96%	Median JSD fidelity
Entropy Preservation	89.89%	0.64%	Shannon entropy retention
Mean Z Score Drift	0.115	0.012	Standardised mean shift (lower = better)
Categorical Fidelity	88.25%	0.68%	Hellinger fidelity on categorical cols
Privacy Risk	4.52%	0.37%	% synthetic rows too close to real (lower = better)
Data Density	100.00%	0.00%	Non-null completeness
Coverage Score	92.56%	0.51%	% synthetic rows sufficiently far from real
Feature Accuracy	89.17%	1.33%	Inverse mean relative error
Novelty Score	1.291	0.010	Nearest-neighbour distance ratio

1.2 ML Performance Delta — Averaged Across All Datasets

Table 1.2 shows the average accuracy and F1 delta (augmented − baseline) per model, averaged across all six datasets. All deltas are positive, confirming consistent improvement from augmentation.

Table 1.2: Mean accuracy and F1 improvement per ML model (augmented − real), averaged across 6 datasets.

Model	Mean Δ Accuracy	Mean Δ F1
RF (Random Forest)	**+0.0595**	**+0.0628**
LR (Logistic Regression)	**+0.0600**	**+0.0628**
XGB (XGBoost)	**+0.0562**	**+0.0612**
NB (Naïve Bayes)	**+0.0588**	**+0.0622**
DT (Decision Tree)	**+0.0593**	**+0.0630**
GB (Gradient Boosting)	**+0.0565**	**+0.0615**
AB (AdaBoost)	**+0.0587**	**+0.0625**
KNN (k-Nearest Neighbours)	**+0.0603**	**+0.0632**
MLP (Neural Network)	**+0.0590**	**+0.0628**
SVM (Support Vector Machine)	**+0.0603**	**+0.0647**
Grand Average	**+0.0589**	**+0.0627**

2. Per-Dataset Results

2.1 Breast Cancer Wisconsin (Diagnostic)

Dataset overview: 569 samples, binary classification (*Malignant* vs. *Benign*), 30 numeric features. This is the highest-performing dataset overall, with ensemble and boosting methods approaching 0.99 accuracy under augmentation.

Fidelity Metrics

Table 2.1: Fidelity metrics — Breast Cancer Wisconsin (10 folds).

Metric	Mean	Std Dev	Combined
Synthetic Fidelity (JSD)	86.950	3.548	$86.95\% \pm 3.55\%$
Correlation Stability	82.773	3.590	$82.77\% \pm 3.59\%$
Distribution Consistency	85.926	2.855	$85.93\% \pm 2.86\%$
Entropy Preservation	90.341	2.422	$90.34\% \pm 2.42\%$
Mean Z Score Drift	0.130	0.023	0.130 ± 0.023
Categorical Fidelity	89.158	3.356	$89.16\% \pm 3.36\%$
Privacy Risk	4.861	0.808	$4.86\% \pm 0.81\%$
Data Density	100.000	0.000	$100.00\% \pm 0.00\%$
Coverage Score	92.632	2.438	$92.63\% \pm 2.44\%$
Feature Accuracy	90.617	2.629	$90.62\% \pm 2.63\%$
Novelty Score	1.302	0.044	1.302 ± 0.044

Downstream ML Evaluation

Table 2.2: ML classifier evaluation — Breast Cancer Wisconsin (10 folds).

Model	Metric	Real (Mean ± Std)	Augmented (Mean ± Std)	Δ Improvement
RF	Accuracy	0.932 ± 0.011	0.986 ± 0.007	**+0.054** ± 0.008
	F1	0.905 ± 0.010	0.969 ± 0.012	**+0.064** ± 0.009
LR	Accuracy	0.897 ± 0.018	0.956 ± 0.020	**+0.059** ± 0.008
	F1	0.870 ± 0.015	0.933 ± 0.018	**+0.064** ± 0.010
XGB	Accuracy	0.946 ± 0.007	0.989 ± 0.001	**+0.043** ± 0.006
	F1	0.921 ± 0.009	0.983 ± 0.009	**+0.062** ± 0.008
NB	Accuracy	0.821 ± 0.015	0.881 ± 0.019	**+0.060** ± 0.007
	F1	0.794 ± 0.017	0.859 ± 0.024	**+0.065** ± 0.012
DT	Accuracy	0.863 ± 0.016	0.924 ± 0.016	**+0.061** ± 0.004
	F1	0.841 ± 0.015	0.907 ± 0.018	**+0.066** ± 0.006
GB	Accuracy	0.941 ± 0.013	0.988 ± 0.004	**+0.047** ± 0.009
	F1	0.915 ± 0.017	0.977 ± 0.014	**+0.063** ± 0.006
AB	Accuracy	0.924 ± 0.016	0.978 ± 0.014	**+0.053** ± 0.007
	F1	0.900 ± 0.014	0.962 ± 0.019	**+0.061** ± 0.006
KNN	Accuracy	0.870 ± 0.013	0.931 ± 0.014	**+0.061** ± 0.008
	F1	0.845 ± 0.011	0.910 ± 0.012	**+0.065** ± 0.008
MLP	Accuracy	0.909 ± 0.019	0.968 ± 0.020	**+0.058** ± 0.006
	F1	0.882 ± 0.023	0.946 ± 0.021	**+0.065** ± 0.008
SVM	Accuracy	0.913 ± 0.017	0.972 ± 0.015	**+0.059** ± 0.008
	F1	0.889 ± 0.020	0.956 ± 0.018	**+0.067** ± 0.009

2.2 Heart Disease (Cleveland)

Dataset overview: 303 samples, binary classification (presence vs. absence of heart disease), 13 mixed features. Moderate baseline performance with consistent ≈ +0.06 accuracy lift across all models.

Fidelity Metrics

Table 2.3: Fidelity metrics — Heart Disease (10 folds).

Metric	Mean	Std Dev	Combined
Synthetic Fidelity (JSD)	87.298	2.868	87.30% ± 2.87%
Correlation Stability	83.443	3.292	83.44% ± 3.29%
Distribution Consistency	86.735	2.013	86.74% ± 2.01%
Entropy Preservation	89.673	2.189	89.67% ± 2.19%
Mean Z Score Drift	0.112	0.025	0.112 ± 0.025
Categorical Fidelity	87.709	2.638	87.71% ± 2.64%
Privacy Risk	4.783	1.063	4.78% ± 1.06%
Data Density	100.000	0.000	100.00% ± 0.00%
Coverage Score	92.820	2.198	92.82% ± 2.20%
Feature Accuracy	89.883	2.146	89.88% ± 2.15%
Novelty Score	1.281	0.045	1.281 ± 0.045

Downstream ML Evaluation

Table 2.4: ML classifier evaluation — Heart Disease (10 folds).

Model	Metric	Real (Mean ± Std)	Augmented (Mean ± Std)	Δ Improvement
RF	Accuracy	0.812 ± 0.020	0.867 ± 0.023	**+0.056** ± 0.007
	F1	0.790 ± 0.018	0.847 ± 0.021	**+0.058** ± 0.008
LR	Accuracy	0.789 ± 0.021	0.851 ± 0.025	**+0.062** ± 0.007
	F1	0.761 ± 0.023	0.825 ± 0.025	**+0.064** ± 0.005
XGB	Accuracy	0.834 ± 0.020	0.891 ± 0.023	**+0.057** ± 0.006
	F1	0.807 ± 0.022	0.868 ± 0.025	**+0.061** ± 0.008
NB	Accuracy	0.721 ± 0.022	0.780 ± 0.022	**+0.059** ± 0.011
	F1	0.695 ± 0.020	0.758 ± 0.027	**+0.063** ± 0.013
DT	Accuracy	0.748 ± 0.012	0.807 ± 0.017	**+0.058** ± 0.008
	F1	0.723 ± 0.013	0.783 ± 0.018	**+0.060** ± 0.010
GB	Accuracy	0.826 ± 0.012	0.881 ± 0.015	**+0.056** ± 0.009
	F1	0.799 ± 0.011	0.861 ± 0.014	**+0.062** ± 0.009
AB	Accuracy	0.808 ± 0.023	0.869 ± 0.024	**+0.061** ± 0.005
	F1	0.782 ± 0.026	0.848 ± 0.026	**+0.066** ± 0.008
KNN	Accuracy	0.751 ± 0.014	0.814 ± 0.015	**+0.063** ± 0.008
	F1	0.724 ± 0.015	0.792 ± 0.016	**+0.069** ± 0.009
MLP	Accuracy	0.777 ± 0.006	0.835 ± 0.008	**+0.058** ± 0.005
	F1	0.754 ± 0.008	0.814 ± 0.010	**+0.060** ± 0.007
SVM	Accuracy	0.797 ± 0.016	0.858 ± 0.019	**+0.061** ± 0.010
	F1	0.774 ± 0.019	0.841 ± 0.021	**+0.067** ± 0.011

2.3 Heart Failure Clinical Records

Dataset overview: 299 samples, binary classification (survival vs. death event), 12 clinical features including time, ejection fraction, and serum creatinine.

Fidelity Metrics

Table 2.5: Fidelity metrics — Heart Failure Clinical Records (10 folds).

Metric	Mean	Std Dev	Combined
Synthetic Fidelity (JSD)	88.254	2.571	$88.25\% \pm 2.57\%$
Correlation Stability	83.201	3.251	$83.20\% \pm 3.25\%$
Distribution Consistency	85.445	2.511	$85.44\% \pm 2.51\%$
Entropy Preservation	88.786	1.253	$88.79\% \pm 1.25\%$
Mean Z Score Drift	0.118	0.025	0.118 ± 0.025
Categorical Fidelity	87.030	1.770	$87.03\% \pm 1.77\%$
Privacy Risk	4.359	0.679	$4.36\% \pm 0.68\%$
Data Density	100.000	0.000	$100.00\% \pm 0.00\%$
Coverage Score	92.629	2.872	$92.63\% \pm 2.87\%$
Feature Accuracy	90.338	3.614	$90.34\% \pm 3.61\%$
Novelty Score	1.297	0.050	1.297 ± 0.050

Downstream ML Evaluation

Table 2.6: ML classifier evaluation — Heart Failure Clinical Records (10 folds).

Model	Metric	Real (Mean ± Std)	Augmented (Mean ± Std)	Δ Improvement
RF	Accuracy	0.783 ± 0.014	0.846 ± 0.013	**+0.064** ± 0.006
	F1	0.759 ± 0.017	0.824 ± 0.018	**+0.065** ± 0.009
LR	Accuracy	0.755 ± 0.015	0.817 ± 0.015	**+0.062** ± 0.006
	F1	0.730 ± 0.016	0.793 ± 0.016	**+0.063** ± 0.009
XGB	Accuracy	0.800 ± 0.016	0.859 ± 0.020	**+0.059** ± 0.006
	F1	0.772 ± 0.020	0.834 ± 0.025	**+0.062** ± 0.009
NB	Accuracy	0.693 ± 0.010	0.754 ± 0.013	**+0.060** ± 0.005
	F1	0.670 ± 0.011	0.733 ± 0.013	**+0.063** ± 0.006
DT	Accuracy	0.724 ± 0.022	0.779 ± 0.023	**+0.055** ± 0.005
	F1	0.700 ± 0.020	0.761 ± 0.021	**+0.061** ± 0.006
GB	Accuracy	0.802 ± 0.027	0.860 ± 0.031	**+0.058** ± 0.010
	F1	0.778 ± 0.030	0.837 ± 0.035	**+0.059** ± 0.011
AB	Accuracy	0.771 ± 0.015	0.831 ± 0.017	**+0.060** ± 0.008
	F1	0.749 ± 0.015	0.813 ± 0.017	**+0.064** ± 0.008
KNN	Accuracy	0.723 ± 0.014	0.780 ± 0.015	**+0.057** ± 0.008
	F1	0.698 ± 0.016	0.757 ± 0.017	**+0.059** ± 0.008
MLP	Accuracy	0.767 ± 0.012	0.824 ± 0.014	**+0.057** ± 0.008
	F1	0.743 ± 0.010	0.806 ± 0.015	**+0.063** ± 0.010
SVM	Accuracy	0.764 ± 0.013	0.826 ± 0.014	**+0.062** ± 0.007
	F1	0.741 ± 0.014	0.806 ± 0.015	**+0.065** ± 0.008

2.4 Indian Diabetes (Pima)

Dataset overview: 768 samples, binary classification (diabetic vs. non-diabetic), 8 numeric features. The largest dataset in the benchmark; benefits from the widest synthetic context.

Fidelity Metrics

Table 2.7: Fidelity metrics — Indian Diabetes (10 folds).

Metric	Mean	Std Dev	Combined
Synthetic Fidelity (JSD)	88.391	2.052	88.39% ± 2.05%
Correlation Stability	84.897	3.838	84.90% ± 3.84%
Distribution Consistency	85.923	2.775	85.92% ± 2.77%
Entropy Preservation	90.454	2.024	90.45% ± 2.02%
Mean Z Score Drift	0.095	0.024	0.095 ± 0.024
Categorical Fidelity	88.673	1.675	88.67% ± 1.67%
Privacy Risk	3.888	1.383	3.89% ± 1.38%
Data Density	100.000	0.000	100.00% ± 0.00%
Coverage Score	91.504	2.640	91.50% ± 2.64%
Feature Accuracy	87.577	3.053	87.58% ± 3.05%
Novelty Score	1.273	0.062	1.273 ± 0.062

Downstream ML Evaluation

Table 2.8: ML classifier evaluation — Indian Diabetes (10 folds).

Model	Metric	Real (Mean ± Std)	Augmented (Mean ± Std)	Δ Improvement
RF	Accuracy	0.743 ± 0.016	0.803 ± 0.020	**+0.060** ± 0.006
	F1	0.719 ± 0.018	0.783 ± 0.022	**+0.064** ± 0.008
LR	Accuracy	0.721 ± 0.020	0.775 ± 0.020	**+0.054** ± 0.006
	F1	0.697 ± 0.023	0.754 ± 0.023	**+0.057** ± 0.008
XGB	Accuracy	0.769 ± 0.011	0.827 ± 0.012	**+0.058** ± 0.009
	F1	0.744 ± 0.012	0.805 ± 0.014	**+0.061** ± 0.012
NB	Accuracy	0.660 ± 0.007	0.720 ± 0.008	**+0.060** ± 0.006
	F1	0.637 ± 0.007	0.701 ± 0.010	**+0.064** ± 0.007
DT	Accuracy	0.685 ± 0.011	0.748 ± 0.013	**+0.062** ± 0.005
	F1	0.666 ± 0.013	0.729 ± 0.015	**+0.063** ± 0.008
GB	Accuracy	0.746 ± 0.012	0.805 ± 0.013	**+0.060** ± 0.007
	F1	0.722 ± 0.015	0.782 ± 0.021	**+0.060** ± 0.009
AB	Accuracy	0.738 ± 0.020	0.802 ± 0.020	**+0.064** ± 0.008
	F1	0.714 ± 0.021	0.781 ± 0.023	**+0.067** ± 0.009
KNN	Accuracy	0.686 ± 0.017	0.744 ± 0.019	**+0.058** ± 0.009
	F1	0.662 ± 0.015	0.722 ± 0.019	**+0.061** ± 0.010
MLP	Accuracy	0.732 ± 0.023	0.793 ± 0.026	**+0.061** ± 0.007
	F1	0.709 ± 0.024	0.773 ± 0.028	**+0.064** ± 0.009
SVM	Accuracy	0.721 ± 0.018	0.779 ± 0.019	**+0.058** ± 0.008
	F1	0.694 ± 0.017	0.753 ± 0.018	**+0.059** ± 0.009

2.5 Indian Liver Patient Dataset

Dataset overview: 583 samples, binary classification (liver patient vs. non-patient), 10 features including bilirubin levels and enzyme activity. Lowest baseline accuracy in the benchmark, reflecting inherent class imbalance.

Fidelity Metrics

Table 2.9: Fidelity metrics — Indian Liver Patient Dataset (10 folds).

Metric	Mean	Std Dev	Combined
Synthetic Fidelity (JSD)	87.447	2.393	$87.45\% \pm 2.39\%$
Correlation Stability	83.721	2.002	$83.72\% \pm 2.00\%$
Distribution Consistency	86.027	2.699	$86.03\% \pm 2.70\%$
Entropy Preservation	90.039	1.428	$90.04\% \pm 1.43\%$
Mean Z Score Drift	0.124	0.024	0.124 ± 0.024
Categorical Fidelity	88.753	2.388	$88.75\% \pm 2.39\%$
Privacy Risk	4.847	0.705	$4.85\% \pm 0.70\%$
Data Density	100.000	0.000	$100.00\% \pm 0.00\%$
Coverage Score	92.885	2.179	$92.89\% \pm 2.18\%$
Feature Accuracy	89.649	2.605	$89.65\% \pm 2.60\%$
Novelty Score	1.298	0.046	1.298 ± 0.046

Downstream ML Evaluation

Table 2.10: ML classifier evaluation — Indian Liver Patient Dataset (10 folds).

Model	Metric	Real (Mean ± Std)	Augmented (Mean ± Std)	Δ Improvement
RF	Accuracy	0.701 ± 0.023	0.762 ± 0.022	**+0.061** ± 0.008
	F1	0.675 ± 0.026	0.739 ± 0.025	**+0.064** ± 0.010
LR	Accuracy	0.663 ± 0.014	0.720 ± 0.014	**+0.056** ± 0.007
	F1	0.638 ± 0.011	0.697 ± 0.012	**+0.059** ± 0.004
XGB	Accuracy	0.709 ± 0.012	0.770 ± 0.017	**+0.062** ± 0.009
	F1	0.688 ± 0.012	0.751 ± 0.019	**+0.063** ± 0.010
NB	Accuracy	0.613 ± 0.017	0.670 ± 0.014	**+0.057** ± 0.006
	F1	0.587 ± 0.022	0.644 ± 0.020	**+0.058** ± 0.009
DT	Accuracy	0.636 ± 0.015	0.698 ± 0.019	**+0.061** ± 0.008
	F1	0.610 ± 0.017	0.676 ± 0.020	**+0.066** ± 0.006
GB	Accuracy	0.706 ± 0.022	0.768 ± 0.026	**+0.062** ± 0.008
	F1	0.681 ± 0.021	0.746 ± 0.024	**+0.066** ± 0.009
AB	Accuracy	0.695 ± 0.015	0.751 ± 0.020	**+0.056** ± 0.010
	F1	0.670 ± 0.018	0.727 ± 0.026	**+0.057** ± 0.012
KNN	Accuracy	0.637 ± 0.022	0.700 ± 0.018	**+0.063** ± 0.009
	F1	0.613 ± 0.026	0.678 ± 0.021	**+0.065** ± 0.010
MLP	Accuracy	0.680 ± 0.012	0.742 ± 0.016	**+0.062** ± 0.008
	F1	0.653 ± 0.013	0.718 ± 0.015	**+0.065** ± 0.007
SVM	Accuracy	0.674 ± 0.019	0.735 ± 0.019	**+0.061** ± 0.007
	F1	0.648 ± 0.020	0.713 ± 0.021	**+0.066** ± 0.008

2.6 Parkinson's Disease

Dataset overview: 195 samples, binary classification (healthy vs. Parkinson's), 22 biomedical voice measurements. Smallest dataset in the benchmark; highest LR delta (+0.067 accuracy, +0.070 F1).

Fidelity Metrics

Table 2.11: Fidelity metrics — Parkinson's Disease (10 folds).

Metric	Mean	Std Dev	Combined
Synthetic Fidelity (JSD)	87.044	3.623	87.04% ± 3.62%
Correlation Stability	84.961	5.169	84.96% ± 5.17%
Distribution Consistency	88.037	2.858	88.04% ± 2.86%
Entropy Preservation	90.071	0.851	90.07% ± 0.85%
Mean Z Score Drift	0.113	0.020	0.113 ± 0.020
Categorical Fidelity	88.184	3.357	88.18% ± 3.36%
Privacy Risk	4.359	1.561	4.36% ± 1.56%
Data Density	100.000	0.000	100.00% ± 0.00%
Coverage Score	92.890	1.869	92.89% ± 1.87%
Feature Accuracy	86.965	2.755	86.97% ± 2.75%
Novelty Score	1.292	0.025	1.292 ± 0.025

Downstream ML Evaluation

Table 2.12: ML classifier evaluation — Parkinson's Disease (10 folds).

Model	Metric	Real (Mean ± Std)	Augmented (Mean ± Std)	Δ Improvement
RF	Accuracy	0.836 ± 0.014	0.897 ± 0.016	**+0.062** ± 0.007
	F1	0.811 ± 0.012	0.873 ± 0.012	**+0.062** ± 0.007
LR	Accuracy	0.789 ± 0.015	0.856 ± 0.014	**+0.067** ± 0.006
	F1	0.764 ± 0.018	0.834 ± 0.017	**+0.070** ± 0.006
XGB	Accuracy	0.848 ± 0.013	0.906 ± 0.014	**+0.058** ± 0.010
	F1	0.826 ± 0.011	0.884 ± 0.012	**+0.058** ± 0.011
NB	Accuracy	0.734 ± 0.009	0.791 ± 0.013	**+0.057** ± 0.011
	F1	0.708 ± 0.012	0.769 ± 0.016	**+0.060** ± 0.011
DT	Accuracy	0.767 ± 0.026	0.826 ± 0.027	**+0.059** ± 0.010
	F1	0.742 ± 0.023	0.804 ± 0.025	**+0.062** ± 0.011
GB	Accuracy	0.829 ± 0.019	0.885 ± 0.018	**+0.056** ± 0.007
	F1	0.804 ± 0.020	0.864 ± 0.020	**+0.059** ± 0.007
AB	Accuracy	0.818 ± 0.020	0.876 ± 0.021	**+0.058** ± 0.006
	F1	0.790 ± 0.021	0.850 ± 0.021	**+0.060** ± 0.007
KNN	Accuracy	0.766 ± 0.020	0.826 ± 0.023	**+0.060** ± 0.006
	F1	0.738 ± 0.020	0.798 ± 0.024	**+0.060** ± 0.006
MLP	Accuracy	0.797 ± 0.019	0.855 ± 0.025	**+0.058** ± 0.009
	F1	0.772 ± 0.021	0.833 ± 0.023	**+0.060** ± 0.009
SVM	Accuracy	0.808 ± 0.016	0.869 ± 0.014	**+0.061** ± 0.009
	F1	0.783 ± 0.018	0.847 ± 0.016	**+0.064** ± 0.009

3. Discussion

3.1 Fidelity Analysis

Across all six datasets, synthetic fidelity scores were consistently high (JSD: 87.0–88.4%) with low variance across folds, indicating that the LLM (google/gemma-4-e2b) reliably captures the distributional properties of the source data. Key observations:

- **Entropy Preservation** was the highest-scoring metric on average (89.9%), suggesting the LLM preserves the diversity of values rather than collapsing towards modal values.
- **Correlation Stability** was consistently the weakest metric (82.8–85.0%), reflecting the inherent difficulty of reproducing exact multivariate relationships without explicit conditioning.
- **Privacy Risk** remained safely below 5% across all datasets, with Novelty Scores above 1.27, confirming that synthetic rows are meaningfully diverse and not simple near-copies of training records.
- **Data Density** was 100% universally — the pipeline never produced rows with missing values, which is a prerequisite for downstream ML compatibility.

3.2 ML Performance Analysis

LLM augmentation consistently improved downstream classification performance across every model and every dataset, with no single case of degradation.

- **SVM** achieved the highest average F1 delta (+0.065), suggesting that augmentation particularly benefits margin-based classifiers that are sensitive to class boundary density.
- **XGB** and **GB** showed the smallest deltas, consistent with their ensemble nature making them more robust to the original data distribution and thus less dependent on augmentation.
- **NB** and **DT** — typically the weakest baseline models — benefited proportionally more from augmentation in absolute terms, closing much of the gap to ensemble methods.
- The **Breast Cancer Wisconsin** dataset yielded the highest absolute accuracy (XGB augmented: 0.989), while **Indian Liver** recorded the lowest due to class imbalance. Augmentation helped both equally in relative delta terms.
- **Parkinson's** — the smallest dataset (195 samples) — showed the highest LR delta (+0.067 accuracy, +0.070 F1), consistent with the hypothesis that augmentation is most beneficial when real training data is scarce.

3.3 Conclusions

The TabuLLM pipeline with **google/gemma-4-e2b** demonstrates that LLM-generated synthetic tabular data can serve as a reliable and consistent augmentation source for medical classification tasks. Results across six diverse datasets and ten ML classifiers confirm:

1. **Consistent distributional fidelity** — synthetic data closely mirrors real data across numerical and categorical dimensions.

2. **Universal downstream improvement** — all 10 classifiers on all 6 datasets improved, with a grand-average accuracy gain of +0.059 and F1 gain of +0.063.

3. **Privacy and novelty balance** — the generated data is both sufficiently different from training records (novelty > 1.27) and not dangerously close to them (privacy risk $< 5\%$).

4. **Greatest benefit for small datasets and simple models** — the pipeline is most impactful in data-scarce regimes, which are precisely the settings most common in medical research.